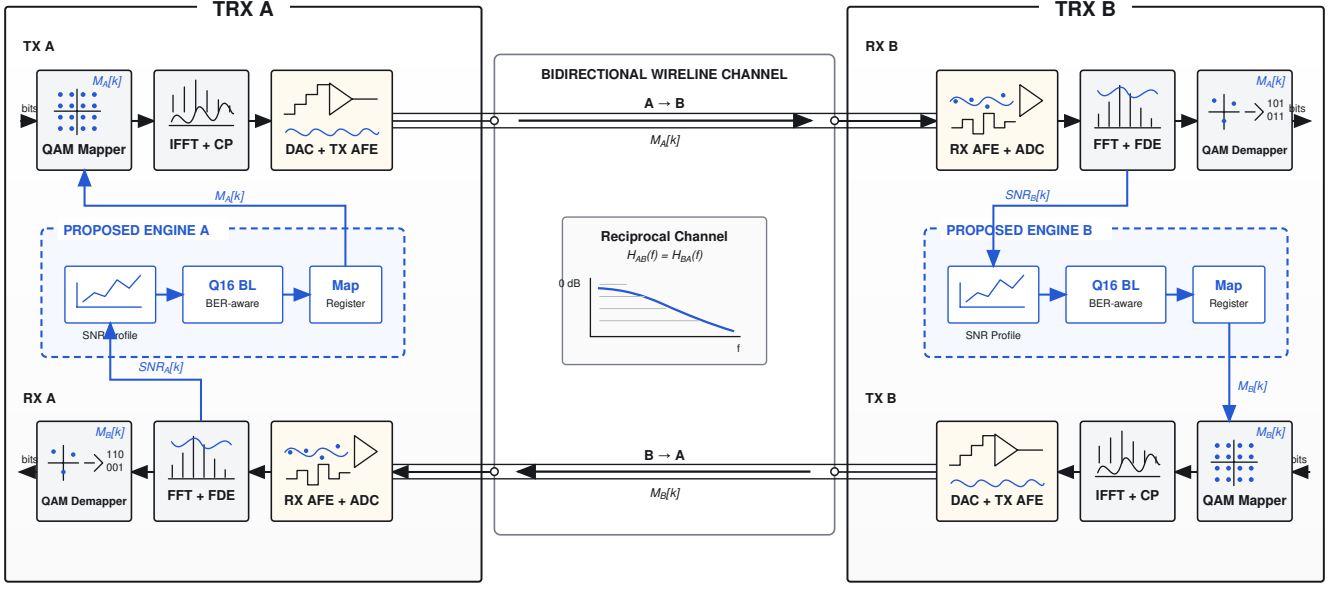
TRX A
TX A
bits
$M_A[k]$
QAM Mapper
IFFT + CP
DAC + TX AFE
$M_A[k]$
PROPOSED ENGINE A
SNR Profile
Q16 BL
BER-aware
Map
Register
$SNR_A[k]$
RX A
bits
$M_B[k]$
110
001
QAM Demapper
FFT + FDE
RX AFE + ADC
BIDIRECTIONAL WIRELINE CHANNEL
A → B
$M_A[k]$
Reciprocal Channel
$H_{AB}(f) = H_{BA}(f)$
0 dB
f
B → A
$M_B[k]$
TRX B
RX B
RX AFE + ADC
FFT + FDE
$M_A[k]$
101
011
QAM Demapper
bits
$SNR_B[k]$
PROPOSED ENGINE B
SNR Profile
Q16 BL
BER-aware
Map
Register
$M_B[k]$
TX B
DAC + TX AFE
IFFT + CP
$M_B[k]$
QAM Mapper
bits